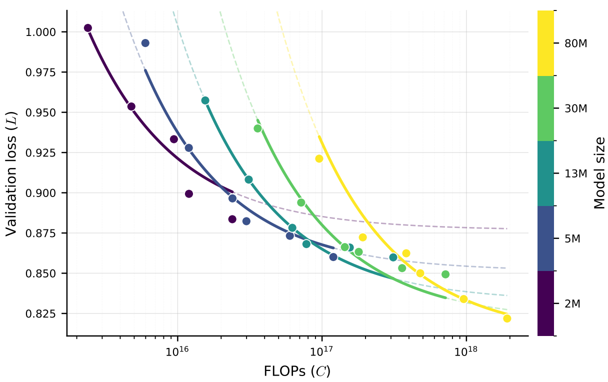
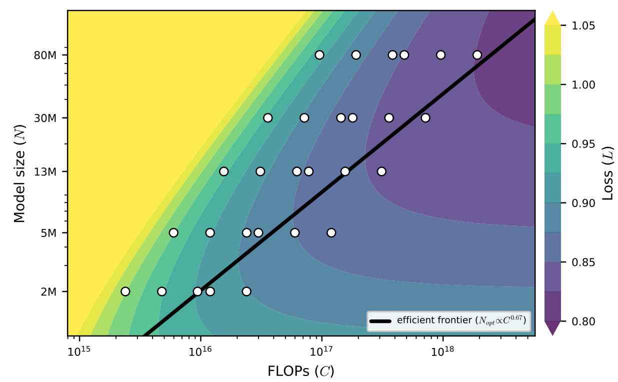


(a) Loss vs. compute, per model size.



(b) Iso-loss contours and efficient frontier.

Scaling behavior decomposed into model size and data. Following Hoffmann et al. (2022) Approach 3, we fit a scaling law that separates the effect of the number of parameters N and the number of training tokens D on the validation loss L. The fit gives $L(N, D) = 0.79 + 16.8\, N^{-0.36} + 1.1 \times 10^5\, D^{-0.72}$. Here we have the parameter exponent $\alpha \approx 0.36$, data exponent $\beta \approx 0.72$, and a compute-optimal model size $N_{\text{opt}} \propto C^{0.67}$. **(a)** Validation loss vs. compute, one curve per model size. **(b)** The fitted loss across compute and model size; the black line marks the compute-optimal model size for each budget.